

Data Analytics Internship

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Domain: Data Analytics

Project: Data-Driven Insights into OLA Ride Bookings using Power BI Dashboard

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Date: 06-05-2026

PHASE 1 — BLUEPRINT DOCUMENT

Project Title

Data-Driven Insights into OLA Ride Bookings using Power BI Dashboard

Project Type

Group Project

1. Problem Statement

Ride-hailing platforms like OLA generate large volumes of booking and ride data. However, without proper analysis, it becomes difficult to:

- Understand booking trends
- Identify cancellation patterns
- Evaluate vehicle performance
- Track revenue and ride efficiency

Traditional reports fail to provide actionable insights, making it challenging for business stakeholders to optimize operations and improve customer satisfaction.

2. Dataset Details

Dataset Name: OLA Bookings Dataset

Source Files Used:

- bookings.csv (Primary dataset)
- ola.sql (Database queries & views)
- ola.pbix (Power BI dashboard)

Type: Structured data (CSV + SQL database)

Description:

The dataset contains ride booking information including ride status, vehicle type, distance, and revenue details.

Key Columns:

- Booking ID
- Booking Status (Success / Cancelled)
- Customer ID
- Vehicle Type
- Ride Distance
- Booking Value (Fare)
- Pickup & Drop Location
- Booking Date / Time

3. Hypothesis

- Higher cancellations reduce overall revenue
- Certain vehicle types generate more revenue than others
- Ride distance impacts booking value significantly
- Peak hours contribute to higher booking volumes
- Customer cancellations are more frequent than driver cancellations

4. Solution Approach

The project follows a structured analytics pipeline:

Step 1: Data Preparation

- Load CSV into SQL database
- Clean missing and inconsistent values
- Create derived columns if needed

Step 2: SQL Analysis (ola.sql)

- Create views such as: Successful bookings, Cancelled rides, Average ride distance per vehicle
- Perform aggregation queries

Step 3: Data Analysis (Python – optional)

- Exploratory Data Analysis (EDA)

- Trend identification
- Pattern recognition

Step 4: Power BI Dashboard (ola.pbix)

- Import cleaned dataset
- Build interactive visuals
- Apply filters and slicers

Step 5: Insights Generation

- Identify business trends
- Highlight performance issues
- Provide recommendation

5. Analysis Plan with Expected Visualizations

Overall Performance

→ KPI Cards

- Total Bookings
- Total Revenue
- Total Distance
- Success Rate

Booking Status Analysis

→ Pie / Donut Chart

- Success vs Cancelled rides

Vehicle Performance

→ Bar Chart

- Revenue by Vehicle Type
- Average Ride Distance

Time-Based Analysis

→ Line Chart

- Bookings over time

- Peak hour demand

Cancellation Analysis

→ Bar Chart

- Cancellation by Customer vs Driver

Distance vs Revenue

→ Scatter Plot

- Relationship between ride distance and fare

Top Insights

→ Table / Bar Chart

- Top performing vehicle
- Most frequent routes

6. Tech Stack

- SQL (MySQL / PostgreSQL) – Data querying (ola.sql)
- Python (Jupyter Notebook) – Optional EDA
- Libraries: Pandas, NumPy
- Visualization: Matplotlib, Seaborn
- Dashboard Tool: Power BI (ola.pbix)

7. Success Criteria

- Dashboard clearly presents key ride metrics
- Visualizations are interactive and insightful
- Business insights are easy to interpret
- Trends like cancellations and peak demand are identified
- Decision-making is supported through data